Problem Chosen	2024 MCM/ICM Summary Sheet	Team Control Number
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Become the Master of Momentum: Insights from Tennis Match Data

Summary

Tennis is a sport full of momentum, which often controls the swings in a tennis match. The power of momentum is sought after by some players and doubted by some coaches. How to quantify momentum? Can momentum play a role in match swings? Are there any features that correlate with momentum? Can we apply momentum to real matches? These are the questions that will be answered in this paper.

To quantify momentum, we construct the **Momentum Swing Model**. Having learned that tennis scoring is not 50/50, we added the probability of a win point on serve to the general formula for calculating momentum. We use this model to present the difference in performance levels between players in a match to capture swings in the match.

To verify the usefulness of momentum for tournaments, we use a variety of statistical methods. After applying autocorrelation function analysis, entropy analysis, and **Kolmogorov-Smirnov test**, we found multiple lines of evidence suggest that momentum plays a role in tennis matches.

To find the features associated with momentum and make predictions, we construct **Momentum Prediction Model**. We analyze and calculate the **turning point** of momentum by the concept of average moving distance. Based on the key event features, we create the **Probability Gain/Loss** method to make predictions about momentum as a way to capture future swings in the match. By constructing the **Momentum Evaluation Model** with the 3S (Search, Summarize, Suggest) process at its core, and summarizing key events of players based on historical match data, we are able to provide them with helpful advice for the future. In addition, we use Djokovic as a case study for practical purposes.

To complement the validation and generalize the model, we use sensitivity analysis and model migration. The results of the sensitivity analysis demonstrate the validity of the models use of constraints. The results of the Momentum Prediciton Model used on womens tennis and table tennis matches show the existence of the models feasibility in generalizing the relevant matches.

The model used in this paper is characterized by strong interpretability, no obvious limitations, and significant results. In order to give tennis coaches an idea of the results of this article, we have written a memo to tennis coaches in which we give some advice on momentum and responding to events during a match.

Keywords: Momentum Swing Model; Probability Model; ACF; Information Entropy; Kolmogorov-Smirnov test; Feature Engineering; Trend Analysis; Sensitivity Analysis

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1 Introduction

Tennis more than any other sport, is a game of momentum. In physics, momentum is defined as "the force that keeps the object moving". But in Tennis, is is the psychological and physical effects of momentum that determine the direction of a match. When momentum is in players' favour, it means they perform better than their oppenents, specifically in terms of winning more points.

We aim to prove that momentum in tennis matches is real by studying the variation of momentum in tennis matches. Then we try to predict the variation of momentum through various correlation indicators to predict swings in tennis match. Finally, by evaluating the previous variation of momentum of a player, we suggest strategies and tactics to help the player to perform better in the next match. In addition, we try to generalize the momentum model and migrate the concept of momentum to other matches to help coaches in other matches to coach their players.

Before formally discussing the power of momentum in tennis match, let's take a brief look at the rules of tennis.

- In a tennis match, the players or teams start on the opposite sides of the net. One player is desginated the server, and the opposing player is the receiver. Service alternates game by game between the two players or teams.
- A *game* consists of a sequence of points played with the same player serving. A game is won by the player who have won at least four points in total and at least two points more the the opponent.
- A *set* consists of a sequence of games played with service alternating between games, ending when a player winning at least six games and at least two games more than the opponent. However, many open tournaments such as Wimbledon now use a tie-breaker mechanism in order to speed up the process. If one player has won six games and the opponent five or six, the first player who wins seven games will win the set. If the score is 6-6 between the two players, then the last game is called a tie-breaker. It is important to note that in a tie-breaker both players take turns serving.
- A *match* consists of a sequence of sets. The outcome is determined through a best of three or five sets system.

In terms of data, `Wimbledon_featured_matches.csv` is the data about every point from all Wimbledon men's matches after the first 2 rounds. `data_dictionary.csv` consists of explanations and examples about `Wimbledon_featured_matches.csv`'s columns.

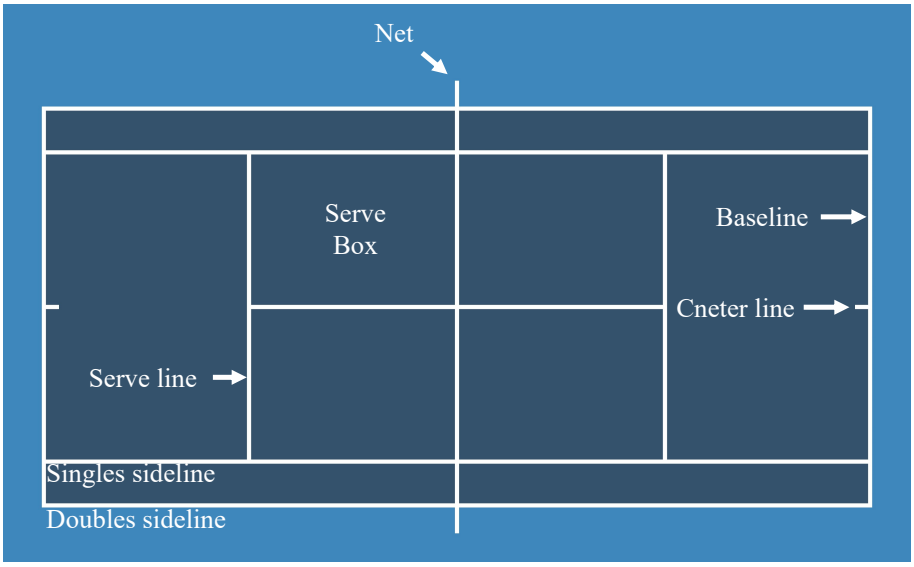


Figure 1: Tennis Court Introduction

Glossory:

Ace: An ace is a winning serve that is not touched by the receiver.

Double Faults: A fault is a serve that falls long or wide of the service box, or does not clear the net. If one player has a fault, they have the chance to begin the second service. If the second service is also a fault, which is called "doublw faults", the recevier wins.

Forehand: The forehand stroke starts on the right side of the body, crosses the body on contact with the ball, and ends on the left side of the body (for right-handed players).

Backhand: The backhand is a stroke that begins on the left side of their body, continues across their body as contact is made with the ball, and ends on the right side of their body (for right-handed players).

1.1 Our Work

Our work is split into six tasks. Each task is based on the results of the previous tasks for further research and calculations.

Task 1: We define a formula for the calculation of momentum and visualize the swing in momentum over the course of a tennis match. The **Momentum Swing Model** is used to show which player is performing better and how much better at a given momentum in time.

Task 2: Base on the Momentum Swing Model, we try to prove the authenticity of momentum in tennis matches from three perspectives: number of points, autocorrelation function analysis, and entropy analysis. Specifically, We simulate the winners and losers of each point randomly on our own and compare them with the data of real games.

Task 3: To predict the momentum swing, we constructed **Momentum Prediction**

Model to make predictions of future momentum from other correlation features. Specifically, we use the average moving distance to calculate the turning point of momentum. We utilize key event features to predict swings in the match by using the Probability Gain/Loss method.

Task 4: In order to help coaches analyze their players and give technical advice, we add the **Momentum Evaluation Model** up the Momentum Prediction Model. Momentum evaluation can analyze a player's history of matches and evaluate the alignment to give technical advice.

Task 5: We generalize the entire Momentum Model to examine and test it on the 2023 Wimbledon men's match dataset, other tennis matches dataset, and other sports matches dataset, respectively.

At last, we write a memo. The memo summarizes the results of our research on Momentum, our advice to coaches, and how to prepare players to respond to swings in play in a tennis match.

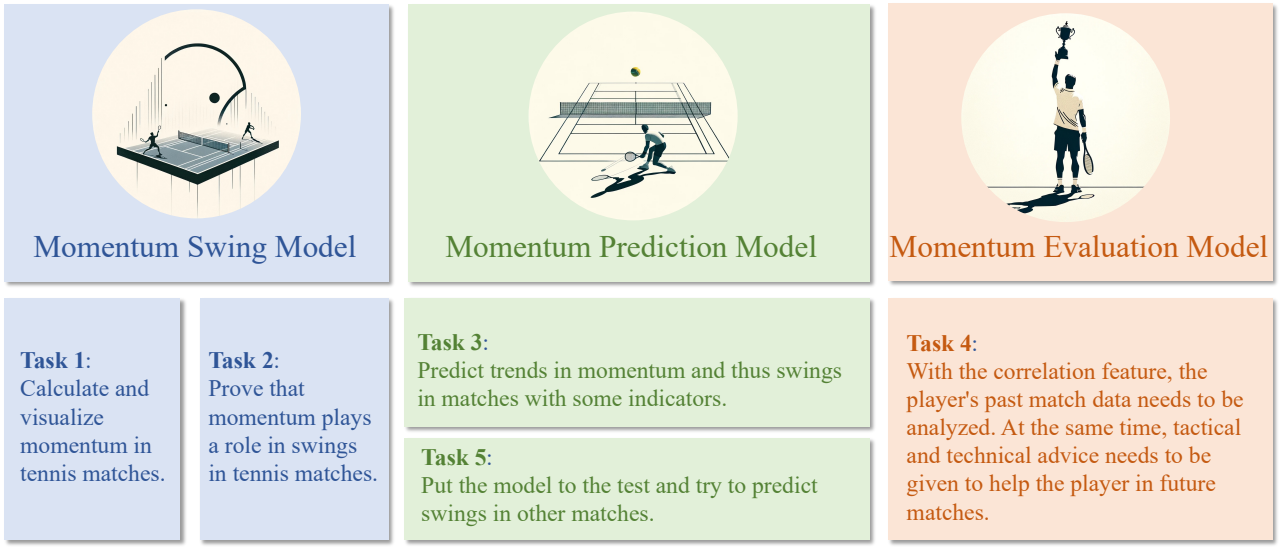


Figure 2: Our Work

1.2 Assumptions and Justifications

- **Assumption:** We assume that tennis players' normal training is enough to provide the stamina to play a full game.
  ↪ ***Justification:*** *Wimbledon is one of four Grand Slam tournaments. Players participating in this international tournament must be professionally trained and have a level of tennis above the amateur level.*

- **Assumption:** We assume that other factors including wind speed, wind direction, sunlight position, etc are not taken into account. The whole match is abstracted as a simple confrontation of players' skills. The serve advantange is the only factor we take count in.

↪ ***Justification:*** *Players who participate in Wimbledon tennis matches are professionally trained. The impressions due to other factors are small enough to be ignored.*

1.3 Notifications

Mathematical Symbols	Explanations
M_i	The momentum at i th point in the whole match.
f_i	The absolute value of momentum change at i th point in the whole match.
$\bar{M}$	Average of the entire momentum sequence.
$P(M_i)$	The probability of occurrence of M_i.
k	The lag of momentum sequence.
m	Random variable of momentum.
H_0	The null hypothesis of Kolmogorov-Smirnov test.
$1_{codition}$	An indicator function, equal to 1 if condition is true and equal to 0 if condition is false.
$F_{\text{sim},p}, F_{\text{real},q}$	The empirical distribution functions that use simulated or real data.
$\mu_{streak}, \sigma_{streak}$	The expectation and standard deviation of the number of streak of points.
$\mu_{move}, \sigma_{move}$	The expectation and standard deviation of the move distences in a match.

2 Model I: Momentum Swing Model

2.1 Assumptions

Momentum is a metric of how much one player performs better than another. But performing well at least has two definitions:

1. Players who perform well feel they have the momentum to likely win the games.
2. Players who perform well win the applause and cheers of the audience.

Since the object of our discussion here is the activity of the tennis players themselves, we choose the first definition.

2.2 Definition of Momentum

Momentum in tennis is a measure of a player's tennis level and mentality. When momentum is in a player's favour, they tend to perform better like a hot knife through butter. At the point where momentum is turning against said players, they tend to perform worse than the opponent as much as they want to play great. The momentum in tennis is specifcally manifested in the winning and losing of each game, as calculated by Wimbledon's official website[1]. This calculation suggests that if one player wins consecutively over a period of time, then it will cause momentum to favor him. If both players play a scorching game with a mix of wins and losses, then momentum will tend to be neutral.

But the fact is that there is no scientific or principled explanation of momentum. Here, we first assume that momentum exists and has an effect on the swings of the tennis match. After that, we will confirm the existence of momentum through mathematical and statistical methods. The calculation shown on Wimbledon's official website suggests that the game is a 50/50 match. But according to our statistics, the probability of serving side winning is 67.3%. Therefore, we consider the formula for momentum as follows:

$$\begin{aligned} &M_0 = 0, \\ &M_i = \begin{cases} M_{i-1} + f_i, & \text{if player1 win} \\ M_{i-1} - f_i, & \text{if player2 win} \end{cases}, \quad i > 0 \end{aligned} \tag{1}$$

where M_i represents the momentum at i th point in the whole match, and f_i represents the absolute value of momentum change at i th point in the whole match. Specifically, f_i is usually 1, but we take into account the fact that it is easy to score on serve. So returning requires more finesse, and as a result player can get more momentum. Specifically, when a player wins a point on return, $f_i = (67.3\%)/(100\% - 67.3\%) \approx 2.06$.

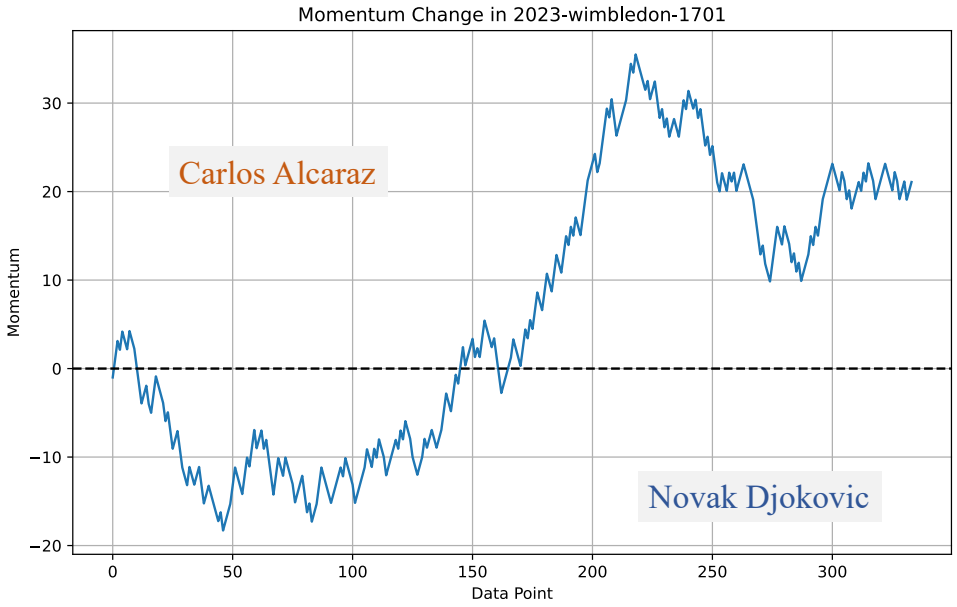


Figure 3: Momentum Swing in the 2023 Wimbledon Gentlemen's Final

Then we apply the formula of momentum to the 2023 Wimbledon Gentlemen's final, where Carlos Alcaraz defeated Novak Djokovic. As shown in Figure 1, Novak Djokovic was the better player at the beginning, and momentum was in his favour. 50 points later, momentum started to rise and fall, indicating that the match was a tight one, but it was still in Novak Djokovic's favour. 125 points later, Carlos Alcaraz was able to hold the advantage for a long time, reversing the whole situation. Although Novak tried to get back into the match after 225 points, it was not to be.

2.3 Proof of Authenticity for Momentum

Intuitively, we think that momentum exists because we are motivated by both familiarity with playing tennis and mental confidence to keep winning points when we win consecutively over a period of time. But we need to prove the authenticity for momentum mathematically.

Here is our solution: We randomly simulate the win/loss data by win/loss probability, and use this data to construct Momentum Swing Model to compare with the real data. On the one hand, we will analyze the overall model difference through line chart, box plot and histogram. On the other hand, we analyze the difference between the simulated data and the real data through autocorrelation function analysis, entropy analysis and Kolmogorov-Smirnov test respectively. Finally, we conclude whether momentum exists or not.

In terms of simulated data, as the probability of winning a point on serve is 67.3%, we used this to perform a random simulation of win/loss data and to simulate a normal match in accordance with the Wimbledon Rules of Tennis. It is worth noting that we took into account the exchange of serves in the deciding set. In terms of real data, we randomly selected 10 match data from the Wimbledon matches dataset. There were 10 matches each of simulated and real data.

Overall Analysis: After simulating and selecting data, we presented the momentum change of simulated data and real data using line graph based on Momentum Swing Model. As shown in Figure 4, we calculated the mean and standard deviation of the real and simulated data in each point separately so as to compare the momentum distributions of the two in a tennis game. We find no significant difference between simulated and real data in the early stages of the game. And in the later stages, the real data start to move in the direction of player 1 as a whole. There are two possible explanations for this phenomenon:

1. The real data has a match that ends after 134 points, so the momentum swing is elevated by the other 9 matches. Because 9 is an odd number, the momentum of different races cannot be completely offset to zero.

2. Since the number of points in the real data averaged 250.3 and the number of points in the simulated data averaged 271.4, we deduce that there is some factor that cause the real game process to be accelerated, making the mean of the real data to deviate by 0.

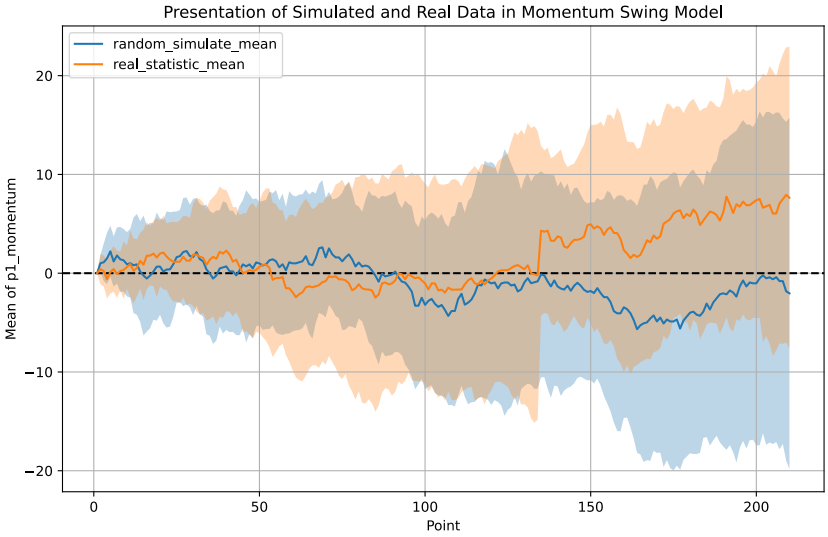
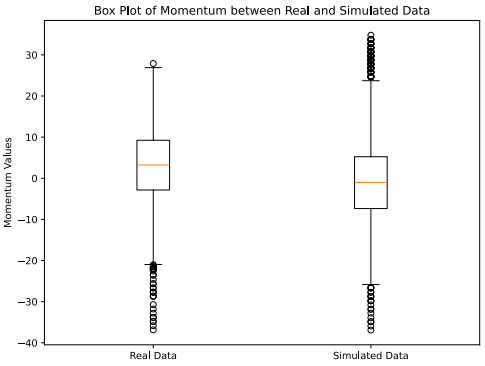
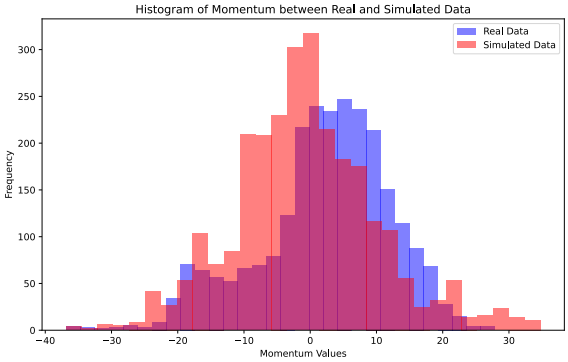


Figure 4: Presentation of Simulated and Real Data in Momentum Swing Model

Since we can not judge the correct explanation, we look at the momentum series as a data distribution and observe the difference between simulated and real data from the perspective of box plots and histograms. As shown in Figure 5, the simulated data shows a normal distribution with a mean of 0. The real data does not have a mean of 0, but has an overall shift in the direction where player 1 is located.



(a) Box Plots



(b) Histograms

Figure 5: Overall Distribution Results

Autocorrelation Function (ACF) Analysis: The Autocorrelation Function (ACF) is a statistical tool used to analyze the degree of similarity between a time series and a lagged version of itself over various time intervals. Mathematically, the ACF at lag k is defined as:

$$ACF(k) = \frac{\sum_{i=1}^{n-k}(M_i - \bar{M})(M_{i+k} - \bar{M})}{\sum_{i=1}^{n}(M_i - \bar{M})^2} \tag{2}$$

where $\bar{M}$ is the average of the entire momentum sequence and n is the number of points in a match.[7]

In our analysis, the ACF graphs are constructed by calculating the mean of autocorrelation coefficients derived from multiple random samples of both real and simulated datasets.

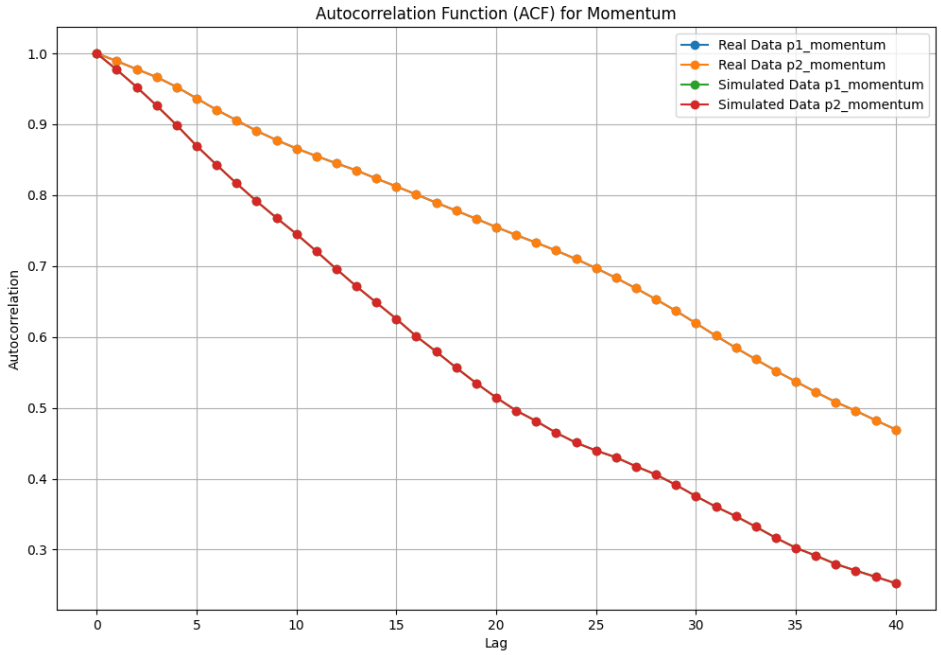


Figure 6: ACF Analysis Results Comparison

The ACF curve for real data typically shows a higher autocorrelation at initial lags, suggesting a non-random pattern where consecutive values are closely related. In contrast, the simulated data often exhibit a more rapidly decreasing ACF curve, indicating a lack of such structured behavior and a more random pattern.

The persistence of higher autocorrelation values in real data indicates underlying systematic factors influencing the momentum feature, which are absent in the simulated data.

Entropy Analysis: Entropy, in information theory, measures the unpredictability or randomness in a dataset. The formula for entropy of momentum M is given by:

$$H(M) = -\sum_{i=1}^{n} P(M_i) \log_2 P(M_i) \tag{3}$$

where $P(M_i)$ is the probability of occurrence of momentum value at i th points in whole match.[5]

The entropy analysis involves calculating the entropy of the momentum feature by averaging the results obtained from multiple random samples of both real and simulated datasets.

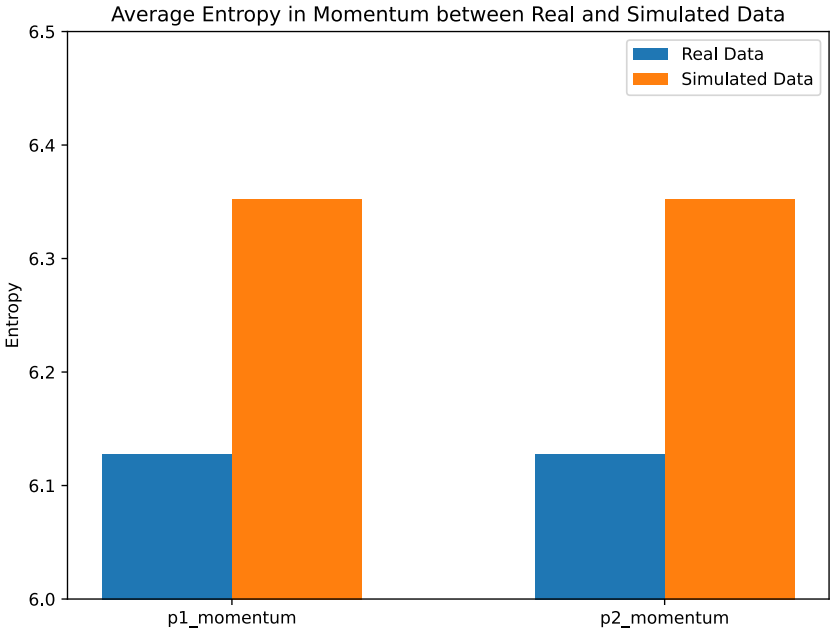


Figure 7: Entropy Analysis Results Comparison

Our findings indicate that the entropy is generally lower for real data compared to simulated data. This lower entropy in real data suggests a higher degree of predictability and structured behavior, as opposed to the higher entropy observed in simulated data, which indicates a greater level of randomness.

This discrepancy in entropy levels between real and simulated data further supports the notion that the momentum variations in real datasets are not purely random, but rather influenced by underlying deterministic factors.

Kolmogorov-Smirnov Test: In statistic, the Kolmogorov-Smirnov test is a nonparametic test of the equality of continuous, one-dimensional probability distributions. It is a common statistical method used to verify whether two data distributions agree. In this paper, our H_0 (null hypothesis) is that real and simulated data follow the same distribution.

We use the empirical distribution function (ECDF) F_n to adopt or reject the null hypothesis, where n is the total number of points in all real or simulated data. The formula of F_n is as follows:

$$F_n(m) = \frac{1}{n}\sum_{i=1}^{n} 1_{M_i \leq m} \tag{4}$$

where m is the random variable of momentum.[6] In this formula, $1_{M_i \leq m}$ is the indicator function, equal to 1 if $M_i \leq m$ and equal to 0 otherwise. The result of empirical distribution function is shown in Figure 8. We find discrepancies between real and simulated data, with simulated data generally having higher empirical distribution function values than real data.

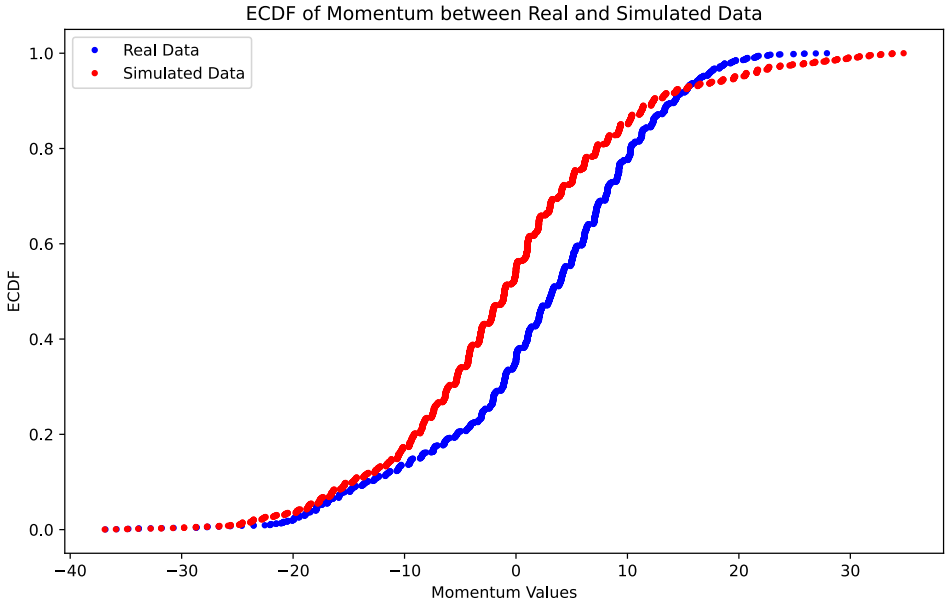


Figure 8: Result of Kolmogorov-Smirnov Test

The calculation method of Kolmogoro-Smirnov statistic is based on ECDF. The formula is as follows:

$$D_{p,q} = \sup_x |F_{sim,p}(m) - F_{real,q}(m)| \tag{5}$$

where $F_{sim,p}(m), F_{real,q}(m)$ are the empirical distribution functions which use simulated data and real data, $D_{p,q}$ is the KolmogoroSmirnov statistic, and $\sup$ is the supremun function. The null hypothesis is rejected at level α if

$$D_{p,q} > \sqrt{-\ln\left(\frac{\alpha}{2}\right) \cdot \frac{1+\frac{q}{p}}{2q}}. \tag{6}$$

The python package `scipy` provides function `ks_2samp` to immediately calculate p value.[2] The null hypothesis can be rejected when $p < \alpha$. We calculate the p value to

be 2.13×10^{-47}, which is much lower than our desired $\alpha : 0.05$. So we reject the null hypothesis and ackowledge momentum's role in swings in tennis matches.

Summary: Through overall analysis, autocorrelation function analysis, entropy analysis and Kolmogorov-Smirnov test, we think we have amply demonstrated that swings in tennis matches are not just random changes, but that there is momentum at play. This important conclusion not only summarizes our current study, but also lays the foundation for the following models.

3 Model II: Momentum Prediction Model

Having come to the conclusion that momentum plays a role in tennis matches, we went on to explore what features influence momentum and how these features can be used to predict swings in matches. In this section, we construct Momentum Prediction Model based on Momentum Swing Model. Unlike Momentum Swing Model, Momentum Prediction Model takes into account some correlation features and predicts the momentum value at each momentum by these features, which can reflect swings in tennis matches.

3.1 Study of Turning Point

As we mentioned in the definition of momentum, players with momentum are like a hot knife through butter and can win multiple points in a row, which is not possible with a simple probabilistic simulations. We therefore infer that the factor providing momentum should be in the time period of the streak of points. Using statistics, we calculated the expectation of the number of streak of points to be 3.11 with a standard deviation of 1.36, denoted by μ_{streak} and σ_{streak}, respectively. We consider the streak of points as a normal distribution for computational simplicity.

In addition, the momentum values, while fluctuating, show a clear overall change as shown in Figure 3. The turning points of the overall change of momentum values is the most important point to reflect that momentum is being affected, and the most difficult point to predict changes in the game. Therefore looking for features associated with turning points is a prerequisite for our predictions to be accurate.

How to determine the turning point is the first question we will study. Since fluctuations in momentum are not uncommon, we need a way to distinguish between local and global fluctuations. We define the distance that momentum moves along the same direction (y-axis positive or negative) as the **move distance**. Using statistics, we calculate the mean and standard deviation of the move distance for each match, donated by μ_{move} and σ_{move}, respectively. We view the distribution of move distances as normal and set the turning point to be the right endpoint of the fold where the move distance exceeds $\mu_{move} + 2 \cdot \sigma_{move}$, since only 5% of moves will exceed it[4], thus isolating most of the local fluctuations.

Intuitively, we believe that there was a turning point likely due to the presence of key events that led to a change in the direction of momentum. So we use a series of important events including "Unforced Error", "Made it to the net", "Won at net", "Double Fault" and

"Ace" at the 2023 Wimbledon Gentlemen's final and compare them to the turning point. As shown in Figure 9, these key events is distributed near the turning points, verifying our ideas.

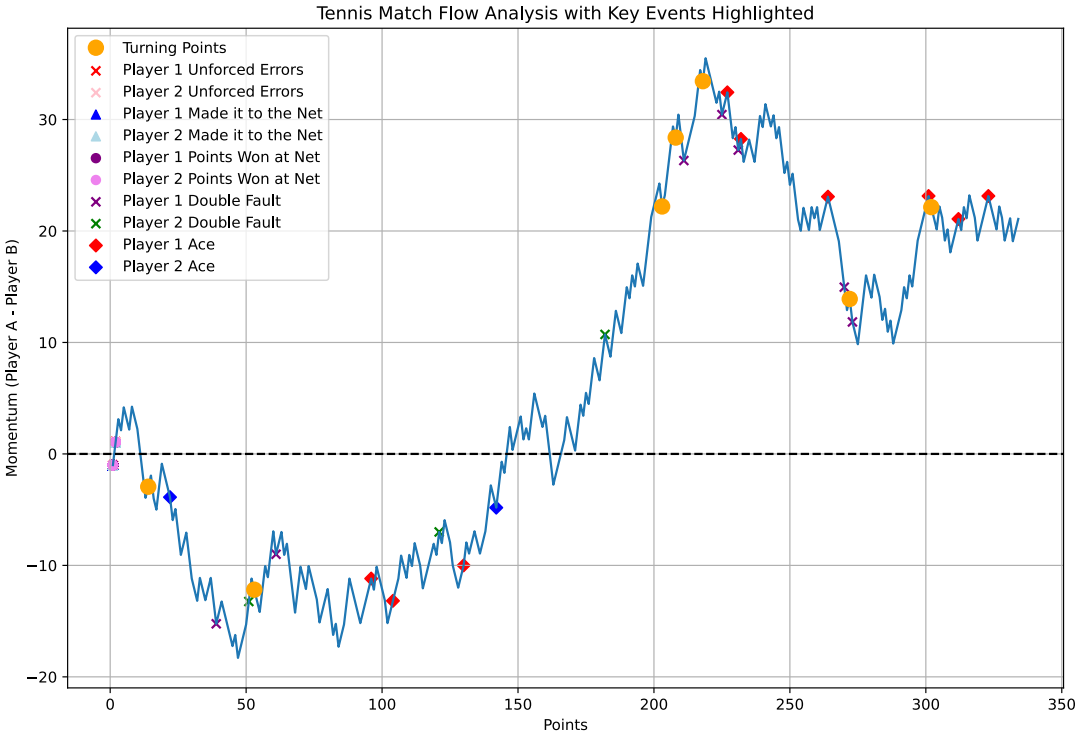


Figure 9: Turning Points and Key Events in 2023 Wimbledon Gentlemen's Final

3.2 Feature Correlation Analysis

Since we found that key event features are related to turning points, we selected all key event features as candidates for further selection. We need to perform a correlation analysis to determine if there is a correlation between features and turning points. Since we want to verify whether the features have an effect on turning points, we want the key events to occur close enough to turning points. So we use the distance to the nearest turning point as a metric of correlation.

Since $\mu_{streak} = 3.11$, $\sigma_{streak} = 1.36$ and the streaks follow a normal distribution, the interval $[\mu_{streak} - 2 \cdot \sigma_{streak}, \mu_{streak} + 2 \cdot \sigma_{streak}]$ covers 95% of the streaks, i.e., $[0.39, 5.83]$. We consider that if the distance between the key event and the nearest turning point is more than 6 (distances can only be whole numbers, rounded here), then this key event data has no effect on the turning point. Since the occurrence of key events is not always correlated with turning points, we set an upper distance limit of 6, consistent with the no-impact points mentioned above, in order to prevent the influence of long distances from uncorrelated points on the final correlation results. The result of correlation analysis

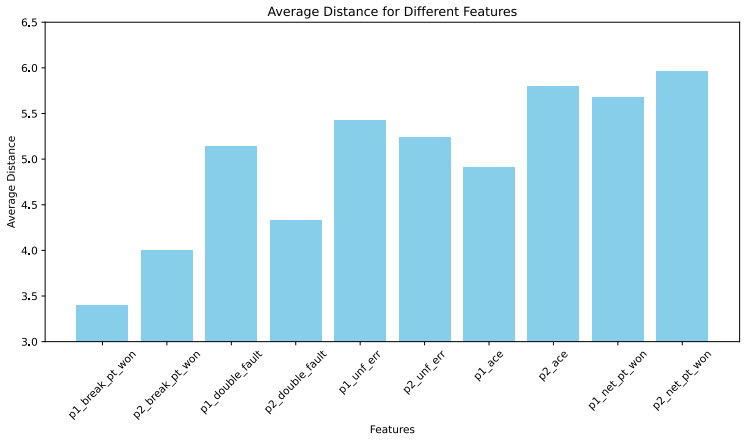


Figure 10: Correlation Analysis Result

result is shown in Figure 10. The sum of the mean distances of the key event features of both players of the game, in order from smallest to largest, are "Won the game the opponet is serving", "Double Fault", "Unforced Error", "Ace" and "Won at net".

Note that to prevent these key event features from being autocorrelated, we plot a heat map of the correlation between features. As shown in Figure 11, the correlation between key event features is not high.

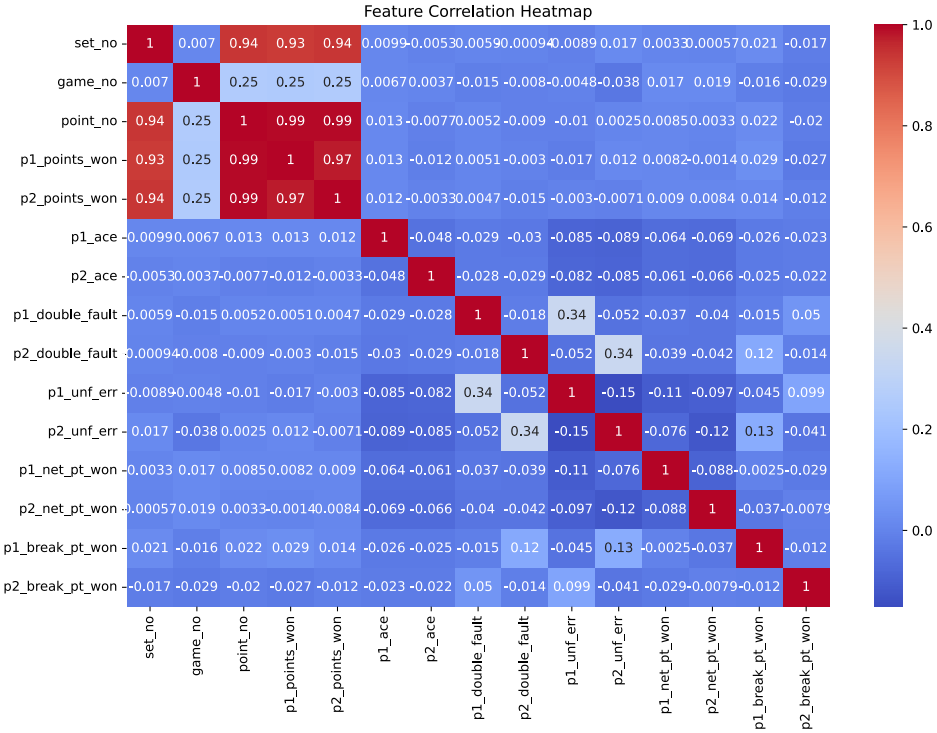


Figure 11: Features Correlation in Heat Map

3.3 Prediction Method: Probability Gain/Loss

After selecting key event features and doing a correlation analysis, we intend to use a probabilistic gain/loss approach to predict swings in tennis matches. The specific method is that we first divide the training set and test set. During the training process, we count the win/loss information within 6 points after the occurrence of ket events. The statistics take into account that the probability of winning point on serve is 67.3%, and we discuss player1 serves and player2 serves separately. After the training process is completed, for each important event feature, we obtain the average gain/loss compared to considering only the probability of winning points on serve.

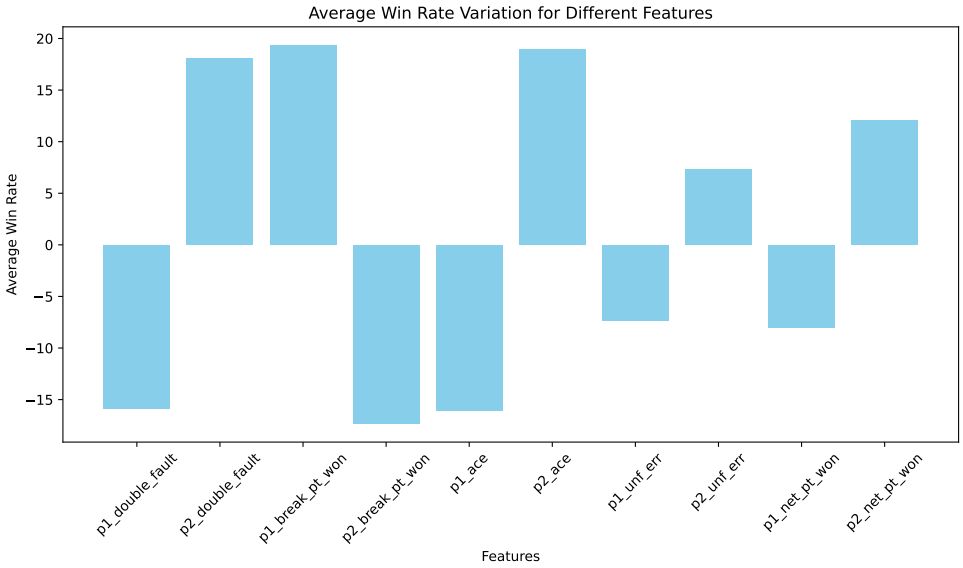


Figure 12: Average Probability Gain/Loss for Key Event Features

As shown in the Figure 12, Those with a probability gain > 0 have a boost to player1 probability of winning points, and those with a probability gain < 0 have a boost to player2 probability of winning points. In the testing phase, for a point victor we want to predict, we first check if a key event occurs in the interval of the first 6 points of this point, and if so probability will gain or lose correspondingly. This allows us to simulate future point victors based on the Momentum Swing Model to predict swings in the tennis match.

Using the 2023 Wimbledon Gentlemen's final as an example, we predict swings in the second half of the match. In 2023 Wimbledon Dataset, there are 31 matches after first two rounds. We use other matched excepte the final as a training set. The remaining final is used as a test set. As shown in Figure 13, the predicted results have the same overall trend compared to the real data, and predict the turning points very well.

It is important to note that we assume that the effects of different key events on the probability of winning or losing at a given point are linearly superimposed. The reason for our assumption is that we previously did a correlation analysis between key event

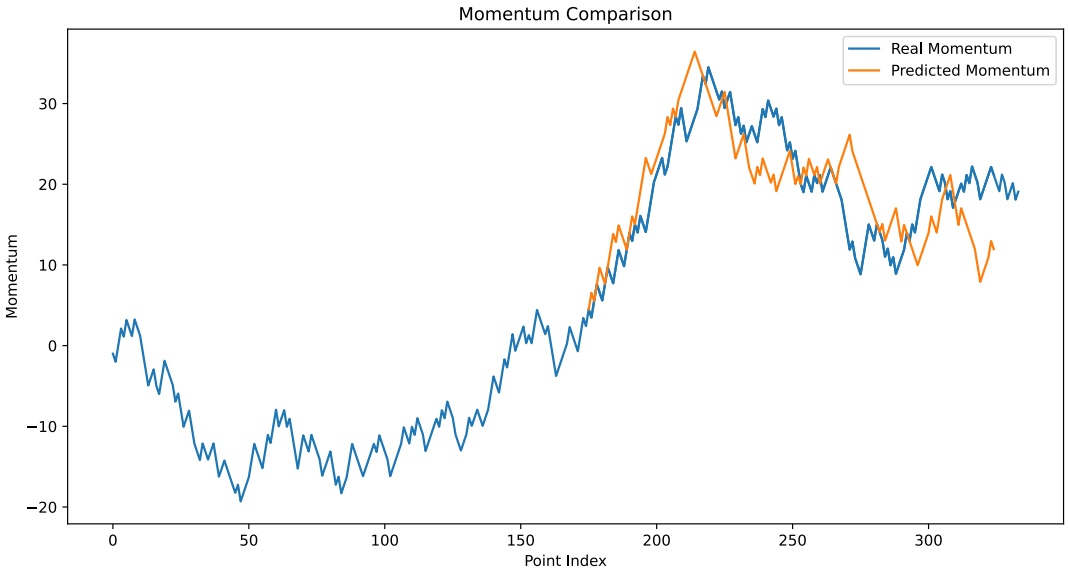


Figure 13: Model Prediction Test Result

features and we got the conclusion that the correlation between features is low. Therefore we believe that we can abstract the multiple correlation effects into a simple linear superposition.

4 Model III: Momentum Evaluation Model

To help coaches give tactic advice based on how players have performed in previous games, we construct Momentum Evaluation Model based on Momentum Prediction Model. We are now able to predict swings in the match by modeling the turning points of the momentum through the Momentum Prediction Model using key event features. We then intend to further apply these key event features to summarize the players' previous matches, analyze the reasons for their losses, and make tactical and technical suggestions for the next match.

4.1 3S Process: Search, Summarize, Suggest

The Momentum Evaluation Model's process consists of three steps: searching for key events, summarizing the enemy situation, and suggesting tactical adjustments.

Search: Given a match, we will find the turning points of swings in the match. Since $\lceil \mu_{streak} + 2 \cdot \sigma_{streak} \rceil = 6$, we search 6 points forward for each turning point to see if there are key events in it.

Summarize: After getting the turning point informations and the corresponding key event informations, we try to count and analyze them. For example, what and how many key events happened in the game before a certain turning point.

Suggest: Based on the results of the analysis, we will analyze the player and give

relevant recommendations. We will players gradually solve the problems presented in the game.

4.2 A Case Study: Djokovic

In 2023 Wimbledon Gentlemen's final, Djokovic ended his undefeated run at Wimbledon since 2013, aborting the remarkable achievement of one of the Grand Slam's all-time great players. But that doesn't mean Djokovic's loss in this match was sudden and unforeseen. We present the entire 2023 Wimbledon Gentlemen match course through Figure 14. Djokovic had four matches before the final, numbered '1316', '1408', '1504', and '1402'. In this case study, we will recapture thes four matches and try to find clues in these match data about Djokovic's failure in the final.

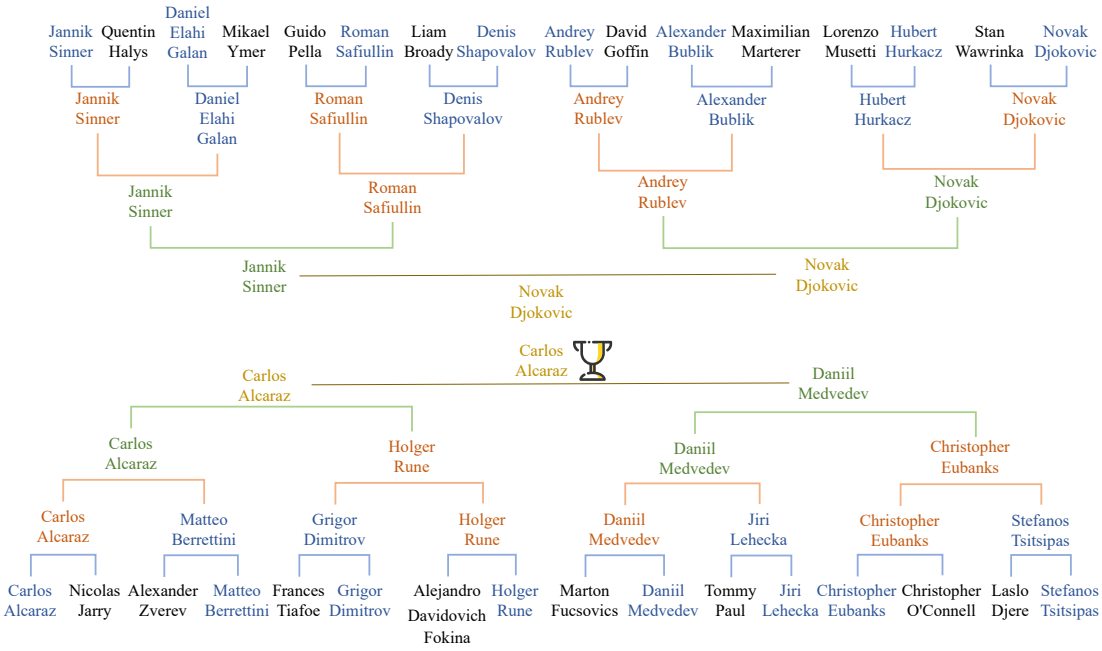


Figure 14: 2023 Wimbledon Gentlemen Match Course

After calculating the turning points of the race and finding the key events in the interval of 6 points forward, we use match data to make the summary graph shown in Figure 15. The blue circles in the figure represent the turning point, and numbers in the center of circles are the point indexes where the turning point occurs. The arrows in the left and right directions on one side of the circles represent the player side of the tendency after the turning points. The texts on either side of the circles are the key events for both players before the turning points.

In the match numbered `1316`, all turning points were in Djokovic's favour. It was an easy match for Djokovic. In the matches numbered `1408` and `1504`, turning points were mostly in Djokovic's favour. However, there were two turning points in favour of the opponent. One of them was when the opponent won 4 points while at the net, while Djokovic had an unforced error. The other turning point was when the opponent won

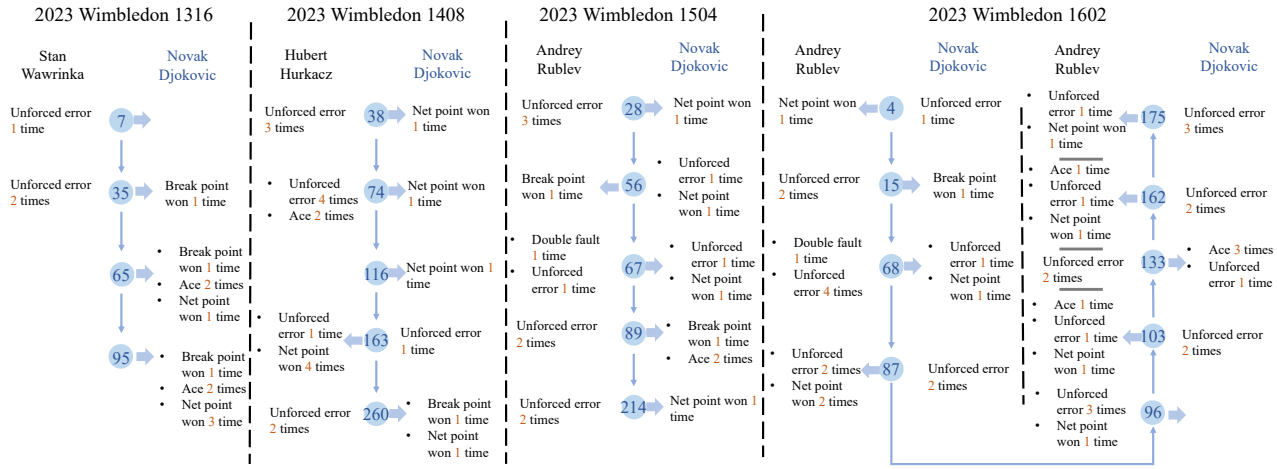


Figure 15: Summary Graph of Djokovic's Four Matches before Final

on Djokovic's serve and Djokovic had an unforced error. In the match numbered 1602, Djokovic and his opponent fought to a scoreless draw, with each holding four turning points in their favour, ending in a close win for Djokovic.

Match ID	double_faults	break_pts	aces	unf_errs	net_pts
2023-wimbledon-1316	0	0	0	5	4
2023-wimbledon-1408	0	0	3	1	4
2023-wimbledon-1504	1	0	0	2	2
2023-wimbledon-1602	0	0	5	6	3

Table 1: Key Events before Turning Points of Djokovic

Match ID	double_faults	break_pts	aces	unf_errs	net_pts
2023-wimbledon-1316	0	0	1	1	0
2023-wimbledon-1408	0	0	1	6	5
2023-wimbledon-1504	0	0	2	4	1
2023-wimbledon-1602	1	0	1	10	9

Table 2: Key Events before Turning Points of Opponents

In terms of Djokovic himself, it only took one unforced error for him to potentially vault a turning point to his opponent, whereas his opponents typically needed 3-4 unforced errors to let their opponents take advantage of a turning point. We think Djokovic is too impatient and too demanding of his own tennis, resulting in a single unforced error that can start tilting momentum in his opponent's favour. We hope that in the next game Djokovic can slow down the pace of the game and focus on the present momentum without being overly influenced by his own mistakes or the opponent's dominance.

5 Validation and Test

After introducing the model, we further analyze the validity, sensitivity and scalability of the model. First we analyze the sensitivity of Momentum Swing Model and Momentum Prediction Model. After that, considering that there might be momentum in other matches, we tried to apply Momentum Prediction Model to other matches including women's tennis matches and table tennis matches.

5.1 Sensitivity Analysis

For the Momentum Swing Model, we consider the probability of winning a point on serve and modified the formula for momentum, but are not sure how the change in probability affected momentum. Therefore, we used the probability of winning a point on serve as the independent variable and the variance of the corresponding momentum as the dependent variable for the sensitivity analysis. The result of sensitivity analysis result of Momentum Swing Model is shown in Figure 16. We can see that Momentum's variance change is more sensitive, so it is reasonable for us to adopt probability of winning a point on serve into Momentum Swing Model.

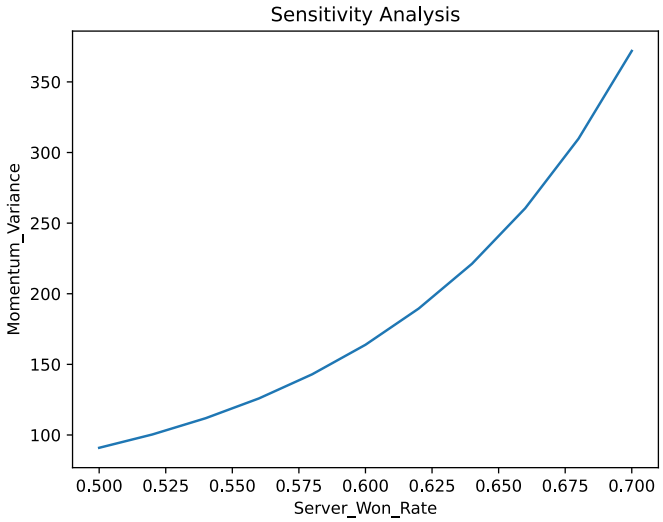


Figure 16: Momentum Swing Model Sensitivity Analysis

For Momentum Prediction Model, since $\lceil \mu_{streak} + 2 \cdot \sigma_{streak} \rceil = 6$, we look 6 points forword the point to be predicted to see if there is a critical event in the prediction to make a prediction. To verify the effect of this correlation threshold on the model results, we performed a sensitivity analysis with the threshold as the independent variable and the predicted momentum mean and standard variance as the response variables. The result of sensitivity analysis is shown in Figure 17. As the correlation threshold increases, the predicted momentum mean shows an increasing and then decreasing trend, and the standard deviation shows a steady decreasing trend. Thus if the correlation threshold changes, the predicted momentum distribution changes. We believe that the correlation

threshold is the one that has an impact on the Momentum Prediction Model results. Momentum Prediction Model has sensitivity to correlation thresholds.

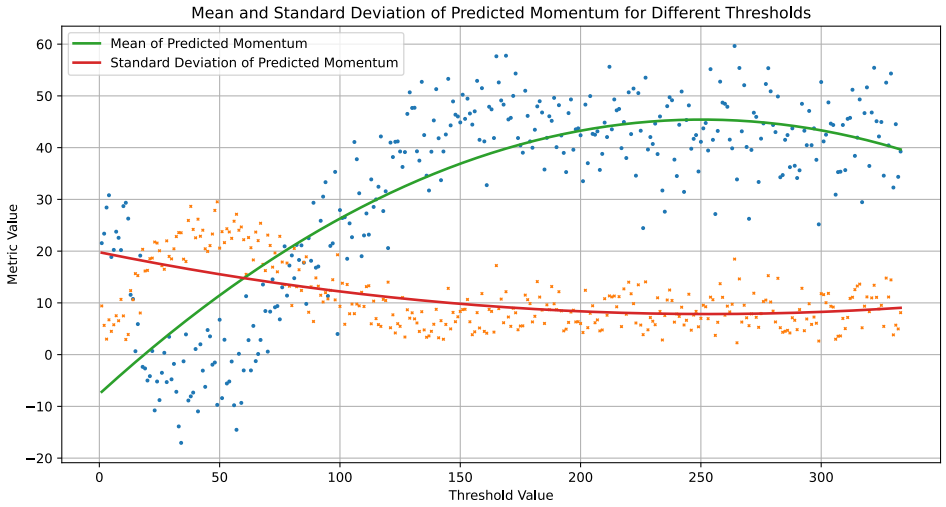


Figure 17: Momentum Prediction Model Sensitivity Analysis

5.2 Application to Women Tennis Matches

In order to verify the scalability of Momentum Swing Model, we try to apply it to other matches. And our primary application target is women's tennis matches. We select the United Cup, Singles witek I. vs Garcia C. match for the date on 6 Jan 2024. We obtained the data for this match from the Sofascore website [3], including each point of the match and the corresponding key events. We used the same probability gain/loss approach to test this match directly after training the Wimbledon dataset.

The result of prediciton is as shown in Figure 18. We find that the prediction is close to the real data, correctly predicting turning points and the direction of the overall trend. Therefore, we believe that there is also momentum in this match and that swings in the match can be predicted by key event features.

5.3 Application to Table Tennis Matches

There are similarities between table tennis and tennis in terms of rules, equipment and facilities. Therefore we chose table tennis matches as a target for further study. Here is the rule of table tennis: In a game, the first team to score 11 points is the winner; after a 10-10 tie, the first team to score 2 more points is the winner. Normally, in official international competitions, the system of four out of seven games is used for individual matches (singles and doubles), and the system of three out of five games is used for singles or doubles in team competitions.

We select Olympic Tournament 2021 Table Tennis Men's Singles Semi-Finals and Finals as our targets. We obtain the data for this match from the Sofascore website [3]. The dataset includes each point of the match and key events features, consist of players'

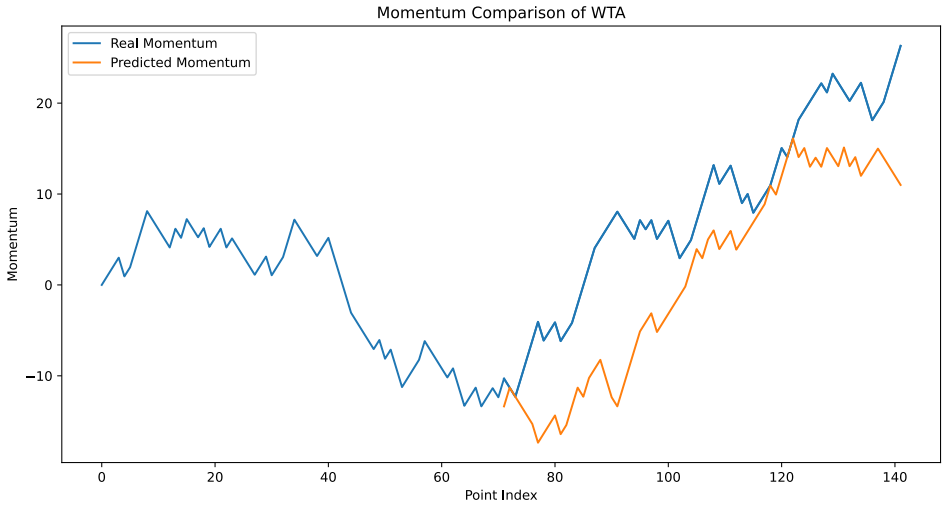


Figure 18: Prediciton Result of Women Tennis Match

max points in a row and players' service errors. Note that we find statistically that the probability of the server winning the point is 50.8%, so we don't take this probability into account in the model as we do in the Momentum Swing Model.

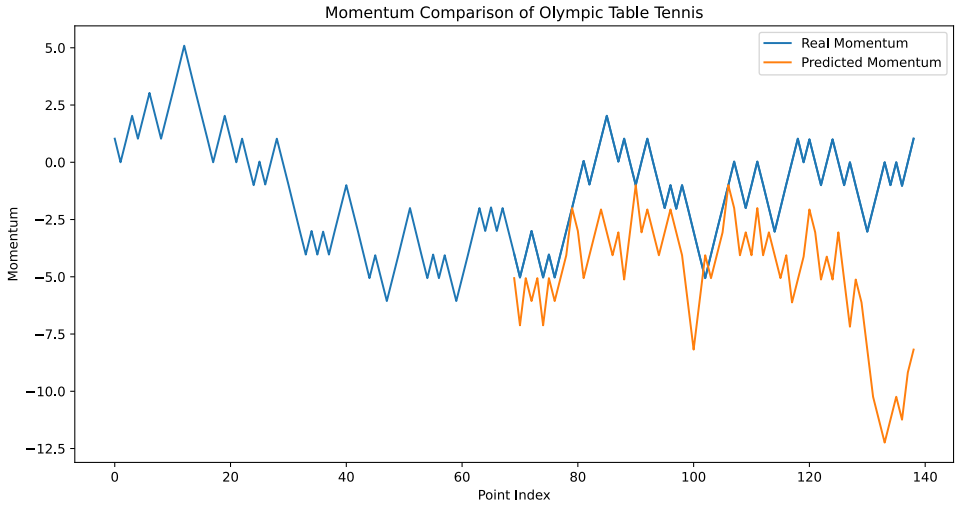


Figure 19: Prediciton Result of Table Tennis Match

Due to the limitation of the dataset, we use the data of 3 matches as the training set and the finals competing for the first and second places as the test set. The prediction results are shown in Figure 19. Due to the small number of our features, the results predicted in the final stage of the match are highly biased. However the predictions of turning points and general trends are similar to the real data. Due to time constraints, we can't conduct the next step in the study.

6 Conclusion

In this paper, we take momentum in tennis game as the research object, and keep on studying momentum deeply from three models, Momentum Swing Model, Momentum Prediction Model and Momentum Evaluation Model. What the Momentum Swing Model does is that given the data from a tennis match, this model allows you to determine which player is playing better and how well. What the Momentum Prediciton Model does is that given the key event features from a tennis match, this model allows you to predict the swings in the match. What the Momentum Evaluation Model does is that given the past match data, this model allows you to evaluate one player's performance and find appropriate tactics and skill adjustments based on the data reflecting the problem.

In addition, we proved the authenticity of momentum in Momentum Swing Model with overall discrepancy, autocorrelation function analysis, entropy analysis and Kolmogorov-Smirnov test to provide theoretical support for three models. We performed a sensitivity analysis of the models to demonstrate the sensitivity of the models to the constant variables we set, verifying that the variables have effect on the models.

Overall, there are advantages and disadvantages of the three models. The three models are highly interpretable, have fewer requirements and restrictions on variables, and are better able to achieve their respective goals and possess good extensibility. However, due to time and resource constraints, we don't make the model more precise and specialized, resulting in the concepts used in the model being shallow and not discussing deeper tennis-like game problems and tennis professional concepts.

If we have enough time to continue discussing this issue in depth, we will read enough books and papers in specialized fields. We will mathematically model the tennis game from a more professional perspective and analyze the concept of momentum in a more professional way. If we have enough resources, we would like to be able to do tennis match experiments to get momentum data from real environments to improve the accuracy of the model.

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Appendices

Appendix A Advice on match momentum and responding to in-match events

Dear, Tennis Coaching
Team

As tennis fans, we appreciate the dedication you give to coach tennis players so that we can see unbelievable tennis matches and feel the momentum of the matches.

We're sure you've heard of the concept of momentum, but some of you may be skeptical that it exists. Momentum is a physical and mental state that controls the change of the match situation. If momentum is in a player's favour, then he has a chance to win consecutive points and take control of the situation.

What we need to acknowledge is that momentum does actually exist, and we have mathematical statistics and simulations that show a discrepancy between the player's game stats and the randomly simulated game stats. That being said, momentum that takes control of the situation exists.

Based on this theory, we have designed three models of momentum and obtained some important results. We would like to share these conclusions and recommendations with you to help tennis players to better improve their performances in matches.

We have worked out that momentum is correlated with key event features, such as ace balls, double faults, unforce errors, and so on. With these features, we are able to predict more accurately the change of the situation of the game and the shift of momentum.

Therefore, these key events are critical to the transformation of momentum. Human beings are emotional creatures and cannot be completely rational. So we sincerely advise you to pay attention to the key events in the game, which may well be a turning point in the situation.

Let's take the example of player Djokovic, who saw his first loss in Wimbledon since 2013 at the 2023 Wimbledon Gentlemen's final. While we feel sorry for Djokovic, we can also analyze the specific reasons for Djokovic's failure. Through the analysis of our model, we find that Djokovic has one unforced error before a momentum shift occurs, while other players often have three or four unforced errors in a row before a momentum shift

occurs. We therefore believe that Djokovic is demanding too much of his tennis and is facing small setbacks for a long time, thus presenting a risk of possible failure.

Therefore, we would like to offer the following advice to help players cope with different game situations.

- When faced with a player's advantage, such as winning a point on the opponent's serve, we recommend that the player does not hesitate to seize the momentum and quickly extend the advantage as a way to gradually improve his momentum. In this way, we can play the role of momentum and keep ourselves winning points.
- When our side loses, we should never be nervous or overthink. In many cases, if we think too much, we will be caught by our opponent's weakness and expand their advantage. We should focus our attention and energy on the present momentum and try our best to play every point.
- Don't be too proud when your opponent is in a losing position. Once you take your opponent lightly, then your opponent will also use it to find your weaknesses and thus turn the situation around. Therefore, it is important to keep a normal mind and try to play every point as well as possible.
- We can't give up even when our opponents are favorable. Because any opportunity is possible before the game is over. We must take it seriously so that we don't let the opportunity go to waste.

The above is our sincere and scientific advice to you regarding the recognition of momentum in the game and the way players deal with different situations. We hope our advice will be helpful to you.

Yours sincerely,

Team#2405234

Report on Use of AI

1. OpenAI ChatGPT (Nov 5, 2023 version, OpenAI DALL·E 3)
   Query1: Draw a minimalist art style picture of two people playing tennis, suitable for use in a thesis.
   Output: Prompt: A minimalistic artistic depiction of two people playing tennis, suitable for inclusion in an academic paper. The scene features a simple tennis court with a net in the middle. One player is on the left side of the court, and the other player is on the right side. Both players are in a dynamic tennis pose, one hitting the ball and the other preparing to return it. The art style is clean and abstract, focusing on the essence of the sport without intricate details. The background is plain to keep the focus on the players and the action.

2. OpenAl ChatGPT (Nov 5, 2023 version, OpenAI DALL·E 3)
   Query1: Draw a picture of a tennis player responding to an incoming ball.
   Output: Prompt: A minimalistic artistic depiction of a tennis player responding to an incoming ball, in the style of the first image provided earlier. The scene features a tennis player in a dynamic pose, capturing the momentum of preparing to hit a returning ball. The player's body language and posture should suggest concentration and agility. The art style is clean and abstract, focusing on the essence of the action without intricate details. The background is plain, highlighting the player and the imminent action of hitting the ball.

3. OpenAI ChatGPT (Nov 5, 2023 version, OpenAI DALL·E 3)
   Query1: Draw a picture of a player holding a trophy.
   Output: Prompt: A minimalistic artistic depiction of a tennis player holding a trophy, in the style of the first image provided earlier. The scene captures the player standing proudly with a trophy held aloft in one hand, symbolizing victory and achievement. The player's posture conveys a sense of triumph and fulfillment. The art style is clean and abstract, maintaining the essence of the momentum without intricate details. The background is plain, emphasizing the player and the trophy, creating a focal point on the success of the athlete.

4. OpenAI ChatGPT (Nov 5, 2023 version)
   Query1: What do you think the "momentum" in tennis match?
   Output: In tennis, "momentum" refers to the psychological and tactical advantage that a player or team gains during a match. It's a concept that's often discussed in sports psychology and is critical in understanding the flow of a tennis match.

   Momentum can shift back and forth between players or teams based on various factors such as winning a crucial point, making a successful shot, capitalizing on an opponent's mistake, or displaying consistent performance over a period of time.

   When a player or team has momentum on their side, they often feel more confident, energized, and motivated. They may play more aggressively, take more risks, and exhibit better focus and determination. Conversely, when momentum shifts against a

player or team, it can lead to frustration, loss of confidence, and a decline in performance.

Momentum in tennis is not just about scoring points but also about the mental and emotional aspect of the game. Players need to manage momentum shifts effectively by staying composed, maintaining a positive mindset, and focusing on their strategy and execution.

Coaches often work with players on techniques to regain momentum if it's lost and to sustain it when they have it. This might involve taking a timeout, changing tactics, staying calm under pressure, or focusing on the process rather than the outcome.

Overall, momentum plays a significant role in the outcome of tennis matches, and understanding how to recognize and control it can be a crucial skill for players at all levels of the game.

5. OpenAI ChatGPT (Nov 5, 2023 version)
Query1: Please tell me the basic rules about tennis match.
Output: Sure! Tennis is a popular racket sport played between two opponents (singles) or two teams of two players each (doubles). Here are the basic rules of tennis:

    1. **Scoring**: Tennis is scored in points, games, sets, and matches. Points are awarded as follows: 15, 30, 40, with the fourth point winning the game. If both players reach 40 (known as deuce), one player must win by two consecutive points to win the game.
    2. **Games**: A game is won by the first player or team to win at least four points and lead by at least two points. The score is announced with the server's score first. If the score reaches 40-40, it's called deuce, and a player must win by two consecutive points.
    3. **Sets**: A set is won by the first player or team to win at least six games and lead by at least two games. If the score reaches 6-6, a tiebreaker is usually played, where the first player or team to reach at least seven points and lead by at least two points wins the tiebreaker and the set.
    4. **Matches**: Matches are typically best-of-three sets for women's matches in Grand Slam tournaments and most other events, while men's matches in Grand Slam tournaments and most other events are best-of-five sets. The winner is the player or team that wins the majority of sets.
    5. **Serving**: The server must stand behind the baseline and serve the ball diagonally into the opponent's service box. The server gets two chances (first and second serve). If they fail to get the serve in the service box, it's called a fault. If a server faults twice in a row, it's a double fault, and the opponent gets a point.
    6. **Court**: The court is divided into two sides by a net. Each side is further divided into service boxes and a baseline. The area between the net and the service line is the forecourt, and the area between the service line and the baseline is the backcourt.

7. **Scoring System**: Love means zero points, 15, 30, and 40 are the points won, and the game is won when a player scores at least four points and is ahead by two.

These are the basic rules of tennis, but there are many more nuances and regulations in the game, including rules regarding faults, lets, foot faults, and more.